

on $\mathbb{R} \times \mathbb{Q}_2/\delta(\mathbb{Z}[\frac{1}{2}])$. The group X_2 constructed in this exercise is a simple example of a *solenoid*.

2.2 Recurrence

One of the central themes in ergodic theory is that of *recurrence*, which is a circle of results concerning how points in measurable dynamical systems return close to themselves under iteration. The first and most important of these is a result due to Poincaré [288] published in 1890; he proved this in the context of a natural invariant measure in the “three-body” problem of planetary orbits, before the creation of abstract measure theory⁽¹⁴⁾. Poincaré recurrence is the pigeon-hole principle for ergodic theory; indeed on a finite measure space it is exactly the pigeon-hole principle.

Theorem 2.11 (Poincaré Recurrence). *Let $T : X \rightarrow X$ be a measure-preserving transformation on a probability space $(X, \mathcal{B}, \mu)$, and let $E \subseteq X$ be a measurable set. Then almost every point $x \in E$ returns to E infinitely often. That is, there exists a measurable set $F \subseteq E$ with $\mu(F) = \mu(E)$ with the property that for every $x \in F$ there exist integers $0 < n_1 < n_2 < \dots$ with $T^{n_i}x \in E$ for all $i \geq 1$.*

PROOF. Let $B = \{x \in E \mid T^n x \notin E \text{ for any } n \geq 1\}$. Then

$$B = E \cap T^{-1}(X \setminus E) \cap T^{-2}(X \setminus E) \cap \dots,$$

so B is measurable. Now, for any $n \geq 1$,

$$T^{-n}B = T^{-n}E \cap T^{-n-1}(X \setminus E) \cap \dots,$$

so the sets $B, T^{-1}B, T^{-2}B, \dots$ are disjoint and all have measure $\mu(B)$ since T preserves μ . Thus $\mu(B) = 0$, so there is a set $F_1 \subseteq E$ with $\mu(F_1) = \mu(E)$ and for which every point of F_1 returns to E at least once under iterates of T . The same argument applied to the transformations T^2, T^3 and so on defines subsets $F_2, F_3, \dots$ of E with $\mu(F_n) = \mu(E)$ and with every point of F_n returning to E under T^n for $n \geq 1$. The set

$$F = \bigcap_{n \geq 1} F_n \subseteq E$$

has $\mu(F) = \mu(E)$, and every point of F returns to E infinitely often. □

Poincaré recurrence is entirely a consequence of the measure space being of finite measure, as shown in the next example.

Example 2.12. The map $T : \mathbb{R} \rightarrow \mathbb{R}$ defined by $T(x) = x + 1$ preserves the Lebesgue measure $m_{\mathbb{R}}$ on $\mathbb{R}$. Just as in Definition 2.1, this means that

$$m_{\mathbb{R}}(T^{-1}A) = m_{\mathbb{R}}(A)$$

for any measurable set $A \subseteq \mathbb{R}$. For any bounded set $E \subseteq \mathbb{R}$ and any $x \in E$, the set

$$\{n \geq 1 \mid T^n x \in E\}$$

is finite. Thus the map T exhibits no recurrence.

The absence of guaranteed recurrence in infinite measure spaces is one of the main reasons why we restrict attention to probability spaces. There is nonetheless a well-developed ergodic theory of transformations preserving an infinite measure, described in the monograph of Aaronson [1].

Theorem 2.11 may be applied when E is a set in some physical system preserving a finite measure that gives E positive measure. In this case it means that almost every orbit of such a dynamical system returns close to its starting point infinitely often (see Exercise 2.2.3(a)). A much deeper property that a dynamical system may have is that *almost every* orbit returns close to *almost every* point infinitely often, and this property is addressed in Sect. 2.3 (specifically, in Proposition 2.14).

Extending recurrence to multiple recurrence (where the images of a set of positive measure at many different future times is shown to have a non-trivial intersection) is the crucial idea behind the ergodic approach to Szemerédi's theorem (Theorem 1.5). This multiple recurrence generalization of Poincaré recurrence will be proved in Chap. 7.

Exercises for Sect. 2.2

Exercise 2.2.1. Prove the following version of Poincaré recurrence with a weaker hypothesis (finite additivity in place of countable additivity for the measure) and with a stronger conclusion (a bound on the return time). Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system with μ only assumed to be a finitely additive measure (see (A.1)), and let $A \in \mathcal{B}$ have $\mu(A) > 0$. Show that there is some positive $n \leq \frac{1}{\mu(A)}$ for which $\mu(A \cap T^{-n}A) > 0$.

Exercise 2.2.2. (a) Use Exercise 2.2.1 to show the following. If $A \subseteq \mathbb{N}$ has positive density, meaning that

$$\mathbf{d}(A) = \lim_{k \rightarrow \infty} \frac{1}{k} |A \cap [1, k]|$$

exists and is positive, prove that there is some $n \geq 1$ with $\overline{\mathbf{d}}(A \cap (A - n)) > 0$ (here $A - n = \{a - n \mid a \in A\}$), where

$$\overline{\mathbf{d}}(B) = \limsup_{k \rightarrow \infty} \frac{1}{k} |B \cap [1, k]|.$$